Task 4: Breast Cancer Diagnosis Prediction

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1 Introduction

This report documents the completion of Task 4, which involves the analysis of the Breast Cancer Wisconsin (Diagnostic) dataset. The objective of this task was to develop a machine learning model capable of classifying tumor diagnoses as either Malignant (M) or Benign (B) based on cytological features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass.

2 Data Preprocessing

The dataset contains 569 instances and 32 attributes. The following preprocessing steps were applied:

- **Data Cleaning:** The 'id' column was removed as it provides no predictive power. The empty column 'Unnamed: 32', an artifact of the CSV format, was also dropped.
- **Encoding:** The target variable 'diagnosis' was encoded numerically: Malignant (M) as 1 and Benign (B) as 0.
- **Splitting:** The data was split into training (80%) and testing (20%) sets to evaluate performance on unseen data.
- **Scaling:** Standard scaling was applied to normalize the feature values, ensuring that features with larger ranges (e.g., Area) did not dominate the objective function of the model.

3 Methodology

For this classification task, a **Logistic Regression** model was selected due to its efficiency and interpretability for binary classification problems.

The model was trained on the scaled training data using the standard optimization algorithm. The random state was fixed to 42 to ensure reproducibility of the results.

4 Results

The model's performance was evaluated using the testing set. The key metrics observed are as follows:

4.1 Accuracy

The model achieved an accuracy of approximately **97.37%**, correctly classifying 111 out of 114 test cases.

4.2 Confusion Matrix

The confusion matrix below illustrates the prediction breakdown:

	Predicted Benign	Predicted Malignant
Actual Benign	70	1
Actual Malignant	2	41

Table 1: Confusion Matrix for Logistic Regression Model

4.3 Classification Report

Precision and recall metrics indicate balanced performance across both classes:

Class	Precision	Recall	F1-Score
Benign (0)	0.97	0.99	0.98
Malignant (1)	0.98	0.95	0.96

Table 2: Detailed Classification Metrics

5 Conclusion

The Logistic Regression model performed exceptionally well on the Breast Cancer diagnostic dataset. With an accuracy exceeding 97%, the model demonstrates high reliability in distinguishing between malignant and benign tumors based on the provided features.

This concludes the requirements for Task 4.